S32K144 EVB

QUICK START GUIDE

REV4.3

APPLIES FOR: S32K144 EVB (SCH-29248 REV B)

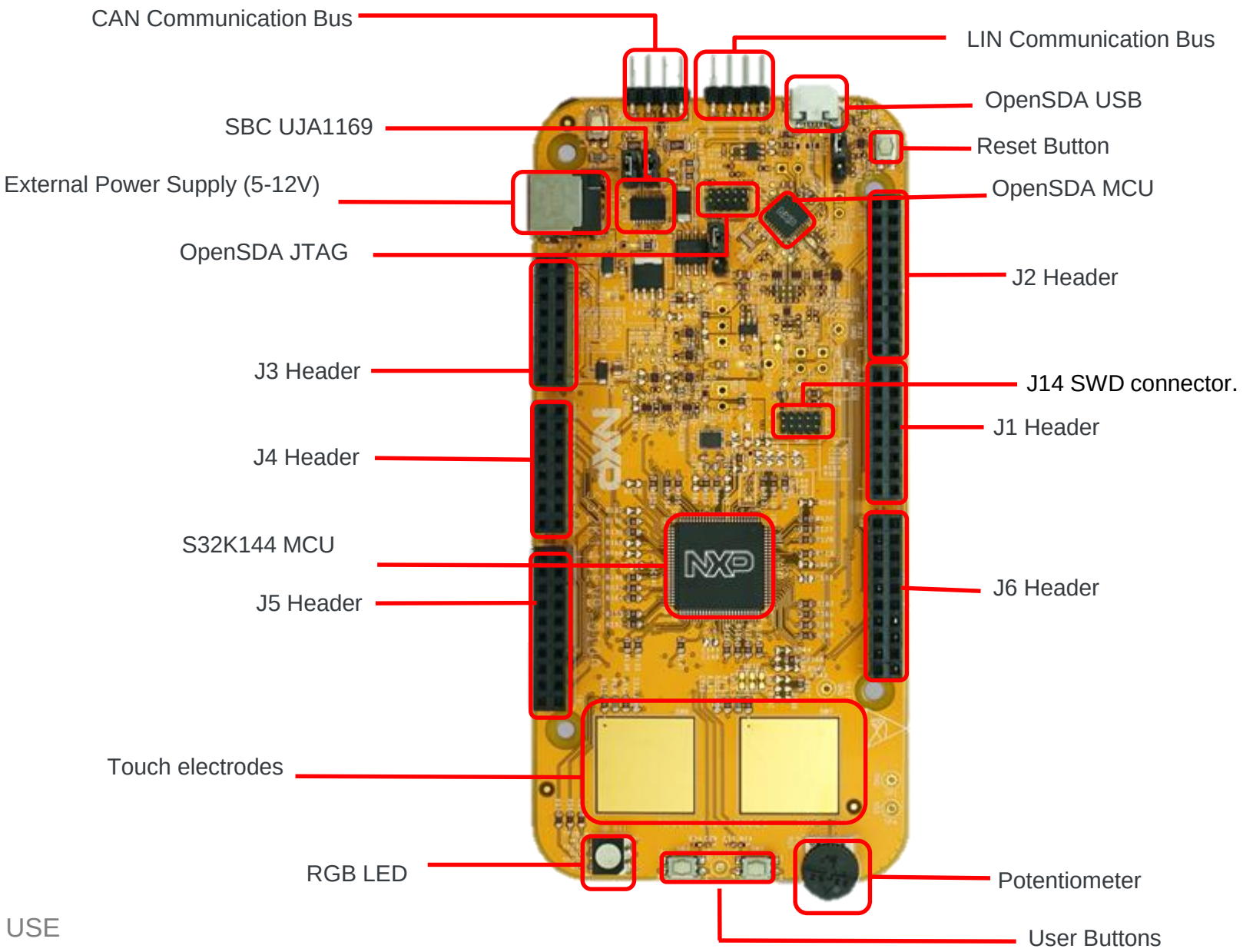
EXTERNAL USE

SECURE CONNECTIONS
FOR A SMARTER WORLD

Contents:

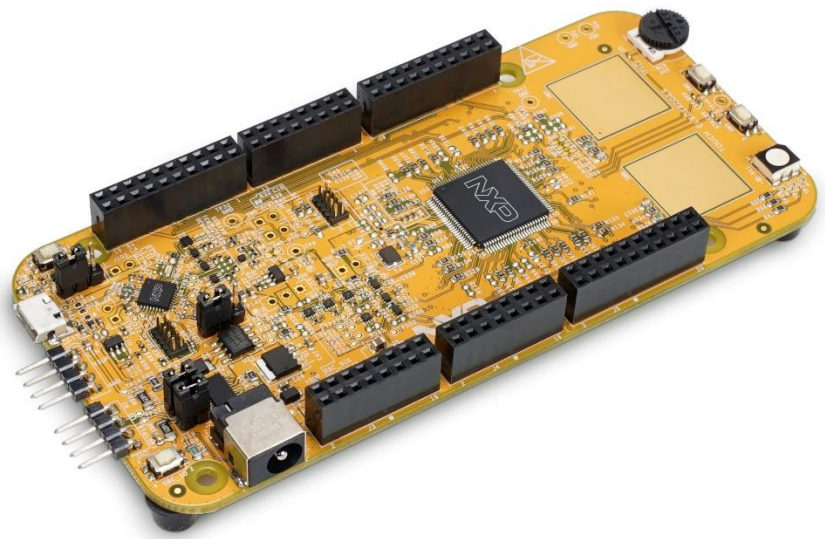
- Get to Know S32K144 EVB
- JumpStart Setup
- JumpStart based on the FreeMASTER tool
- Introduction to OpenSDA
- Creating a new S32DS project for S32K144
- S32DS Debug basics
- Create a P&E debug configuration

Get to know S32K144-EVB



S32K144 EVB Features:

- Supports **S32K144 100LQFP**
- Small form factor size supports up to 6” x 4”
- Arduino™ UNO footprint-compatible with expansion “shield” support
- Integrated open-standard serial and debug adapter (OpenSDA) with support for several industry-standard debug interfaces
- Easy access to the MCU I/O header pins for prototyping
- On-chip connectivity for CAN, LIN, UART/SCI.
- SBC UJA1169 and LIN phy TJA1027
- Potentiometer for precise voltage and analog measurement
- RGB LED
- Two push-button switches (SW2 and SW3) and two touch electrodes
- Flexible power supply options
  - microUSB or
  - external 12V power supply



Header/Pinout Mapping for S32K144

J3

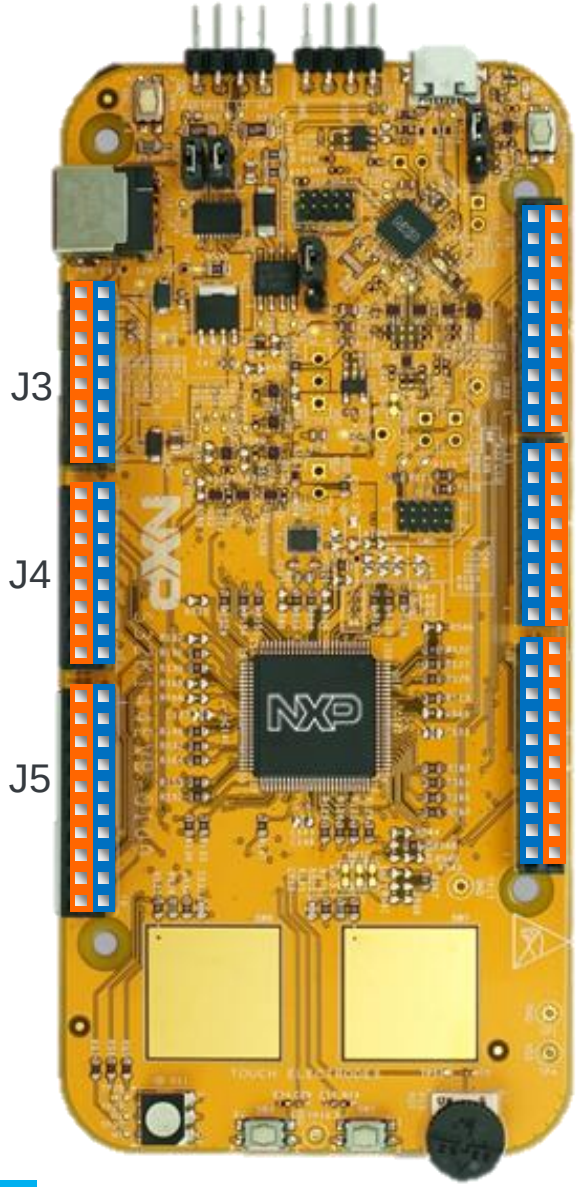
PIN	PORT	FUNCTION	PIN	PORT	FUNCTION
J3-02	PTB6*	GPIO	J3-01		VIN
J3-04	PTB7*	GPIO	J3-03		IOREF
J3-06	PTE0	GPIO	J3-05	PTA5	RESET
J3-08	PTE9	GPIO	J3-07		3V3
J3-10	PTC5	GPIO	J3-09		5V
J3-12	PTC4	GPIO	J3-11		GND
J3-14	PTA10	GPIO	J3-13		GND
J3-16	PTA4	GPIO	J3-15		VIN

J4

PIN	PORT	FUNCTION	PIN	PORT	FUNCTION
J4-02	PTC7	GPIO	J4-01	PTD4	ADC0
J4-04	PTC6	GPIO	J4-03	PTB12	ADC1
J4-06	PTB17	GPIO	J4-05	PTB0	ADC2
J4-08	PTB14	GPIO	J4-07	PTB1	ADC3
J4-10	PTB15	GPIO	J4-09	PTA6/PTE11/PTA2	ADC4
J4-12	PTB16	GPIO	J4-11	PTC0/PTE10/PTA3	ADC5
J4-14	PTC14	GPIO	J4-13	PTE2	ADC6
J4-16	PTC3	GPIO	J4-15	PTE6	ADC7

J5

PIN	PORT	FUNCTION	PIN	PORT	FUNCTION
J5-02	PTE16	GPIO	J5-01	PTA15/PTD11	ADC8
J5-04	PTE15	GPIO	J5-03	PTA16/PTD10	ADC9
J5-06	PTE14	GPIO	J5-05	PTA1	ADC10
J5-08	PTE13	GPIO	J5-07	PTA0	ADC11
J5-10		VDD	J5-09	PTA7	ADC12
J5-12		GND	J5-11	PTB13	ADC13
J5-14	PTE1	GPIO	J5-13	PTC1	ADC14
J5-16	PTD7	GPIO	J5-15	PTC2	ADC15
J5-18	PTD6	GPIO	J5-17	NC	GPIO
J5-20	PTC15	GPIO	J5-19	NC	N/A



J2

PIN	PORT	FUNCTION	PIN	PORT	FUNCTION
J2-19	PTE10/PTA3	D15/I2C_CLK	J2-20	NC	GPIO
J2-17	PTE11/PTA2	D14/I2C_SDA	J2-18	NC	GPIO
J2-15		ANALOGUE REF	J2-16	PTA14	GPIO
J2-13		GND	J2-14	PTE7	GPIO
J2-11	PTB2	D13/SPI_SCK	J2-12	PTC13	GPIO
J2-09	PTB3	D12/SPI_SIN	J2-10	PTC12	GPIO
J2-07	PTB4	D11/SPI_SOUT	J2-08	PTE8	GPIO
J2-05	PTB5	D10/SPI_CS	J2-06	PTD0	GPIO
J2-03	PTD14	D9/PWM	J2-04	PTD16	GPIO
J2-01	PTD13	D8/PWM	J2-02	PTD15	GPIO

J1

PIN	PORT	FUNCTION	PIN	PORT	FUNCTION
J1-15	PTC11/PTE8	D7	J1-16	PTE3	GPIO
J1-13	PTC10/PTC3	D6	J1-14	PTD3	GPIO
J1-11	PTB11	D5	J1-12	PTD5	GPIO
J1-09	PTB10	D4	J1-10	PTD12	GPIO
J1-07	PTB9	D3	J1-08	PTD11	GPIO
J1-05	PTB8	D2	J1-06	PTD10	GPIO
J1-03	PTA3	D1	J1-04	PTA17	GPIO
J1-01	PTA2	D0	J1-02	PTA11	GPIO

J6

PIN	PORT	FUNCTION	PIN	PORT	FUNCTION
J6-19	PTA9	D14	J6-20	PTE4	GPIO
J6-17	PTA8	D15	J6-18	PTE5	GPIO
J6-15	PTE12	D16	J6-16	PTA12	GPIO
J6-13	PTD17	D17	J6-14	PTA13	GPIO
J6-11	PTC9	D18	J6-12		GND
J6-09	PTC8	D19	J6-10		VDD
J6-07	PTD8	D20	J6-08	PTC16	GPIO
J6-05	PTD9	D21	J6-06	PTC17	GPIO
J6-03	PTD2	GPIO	J6-04	PTD3	GPIO
J6-01	PTD0	GPIO	J6-02	PTD1	GPIO

Arduino compatible pins

NXP pins

*0ohm resistor is not connected

Jumper Settings

Jumper	Configuration	Description
J104	1-2	Reset signal to OpenSDA, use to enter into OpenSDA Bootloader mode
	2-3 (Default)	Reset signal direct to the MCU, use to reset S32K144.
J107	1-2	S32K144 powered by 12V power source.
	2-3 (Default)	S32K144 powered by USB micro connector.
J109/J108	1-2 (Default)	Removes CAN termination resistor

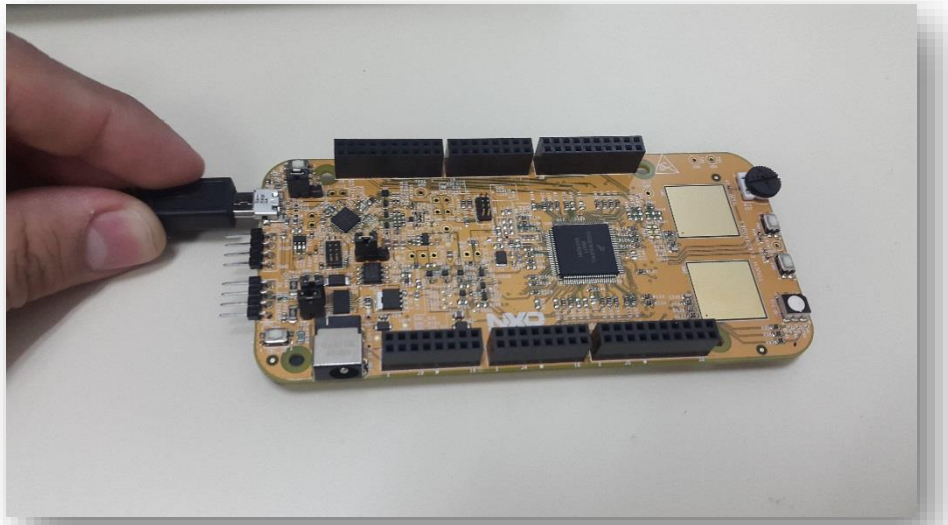
HMI mapping

Component	S32K144
Red LED	PTD15 (FTM0 CH0)
Blue LED	PTD0(FTM0 CH2)
Green LED	PTD16(FTM0 CH1)
Potentiometer	PTC14 (ADC0_SE12)
SW2	PTC12
SW3	PTC13
OpenSDA UART TX	PTC7(LPUART1_TX)
OpenSDA UART RX	PTC6(LPUART1_RX)
CAN TX	PTE5(CAN0_TX)
CAN RX	PTE4 (CAN0_RX)
LIN TX	PTD7(LPUART2_TX)
LIN RX	PTD6 (LPUART2_RX)
SBC_SCK	PTB14 (LPSPI1_SCK)
SBC_MISO	PTB15(LPSPI1_SIN)
SBC_MOSI	PTB16(LPSPI1_SOUT)
SBC_CS	PTB17(LPSPI1_PCS3)

S32K144 EVB JUMPSTART

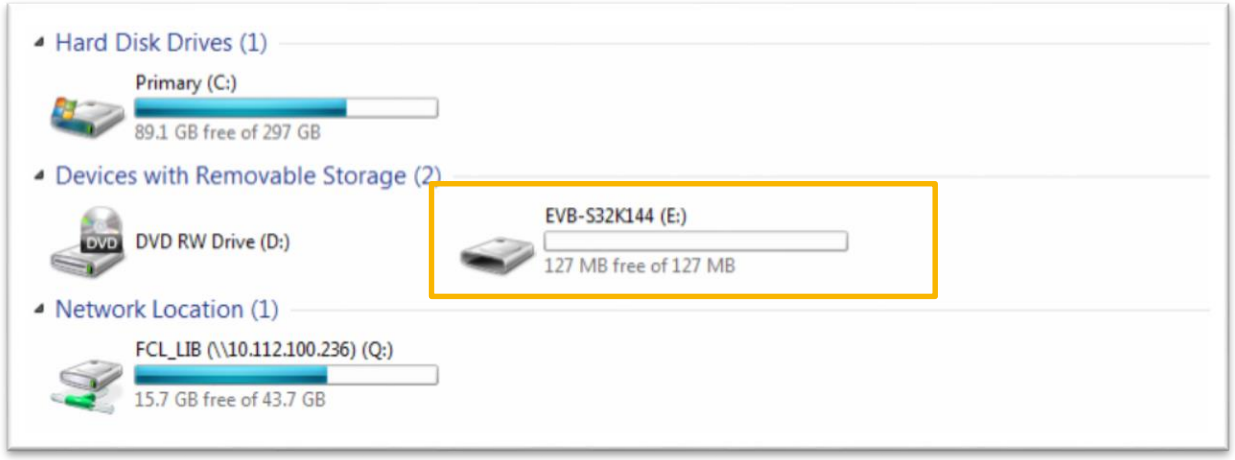
Step 1: Power up the Board – EVB Power Supplies

- The S32K144-EVB evaluation board powers from a USB or external 12V power supply. By default USB power is enabled with J107 (check slide 5)
- Connect the USB cable to a PC using supplied USB cable .
- Connect other end of USB cable (microUSB) to mini-B port on FRDM-KEA at J7
- Allow the PC to automatically configure the USB drivers if needed
- Debug is done using OpenSDA through J7



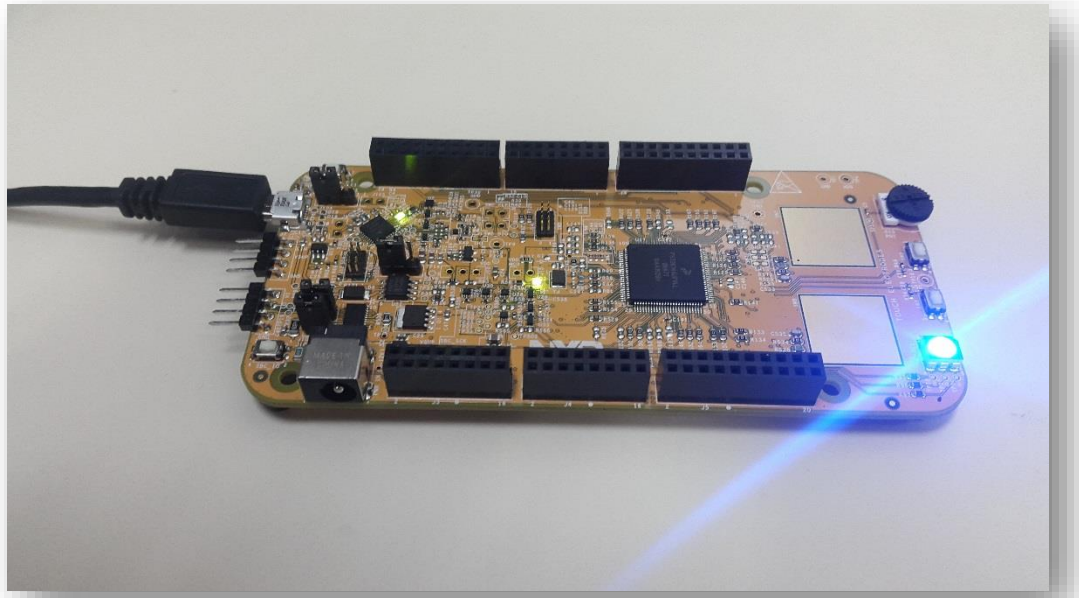
Step 1: Power up the Board – Is it powered on correctly?

- When powered through USB, LEDs D2 and D3 should light green
- Once the board is recognized, it should appear as a mass storage device in your PC with the name EVB-S32K144.



Step 1: Power up the Board – Is it powered on correctly?

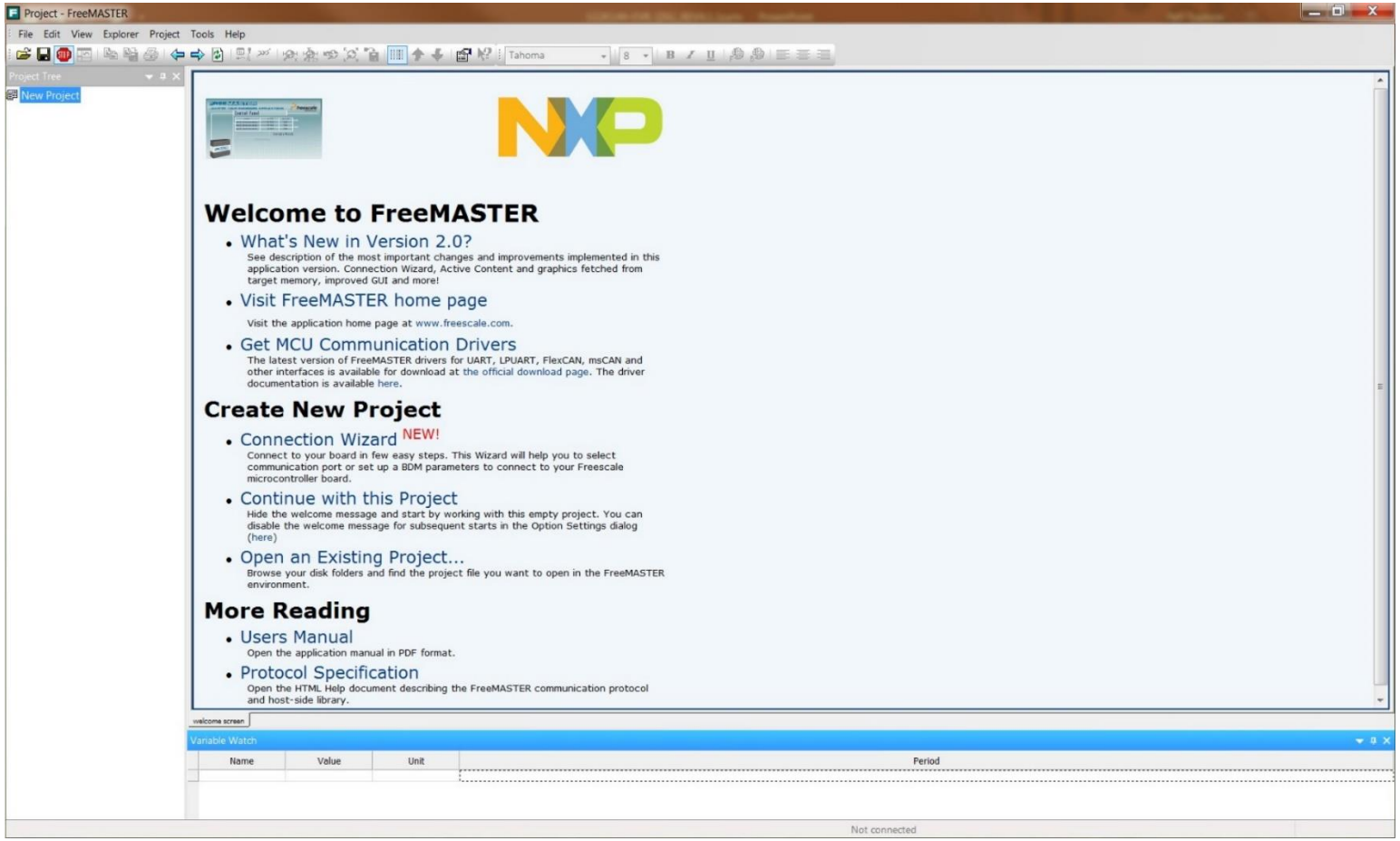
- Board is preloaded with a software, in which the red, blue and green leds will toggle at different rates.



S32K144 EVB JUMPSTART BASED ON THE FREEMASTER TOOL

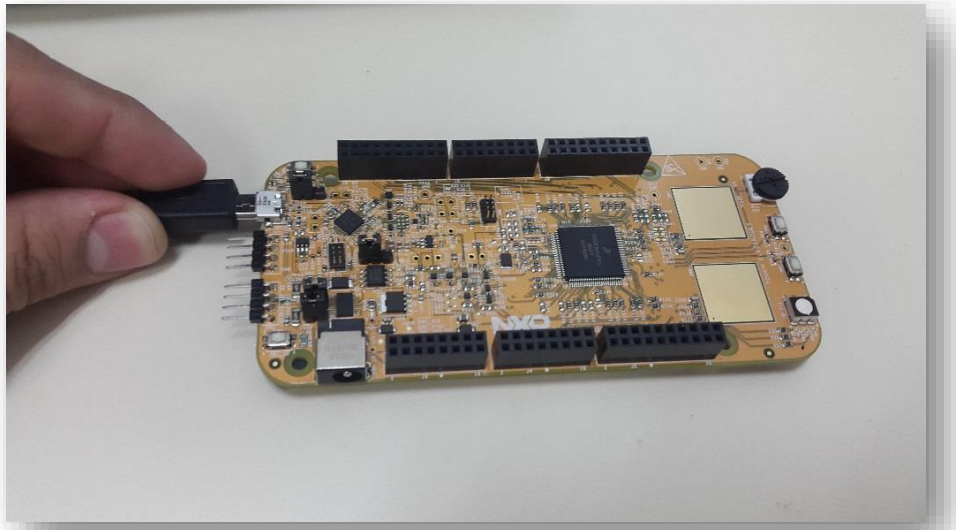
Install the FreeMASTER tool

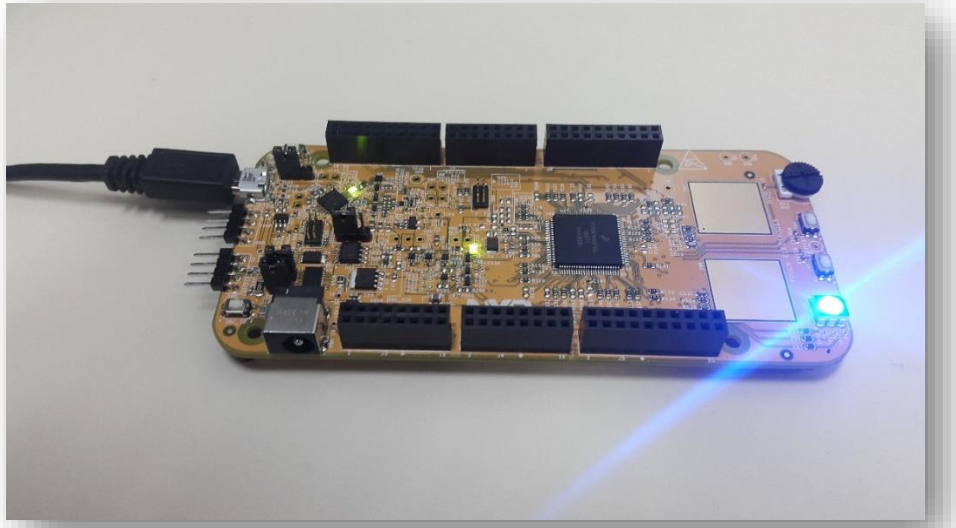
- Download and install the FreeMASTER PC application www.nxp.com/FreeMASTER .
- Open the FreeMASTER application on your PC. You should see Welcome page:



Power up the EVB board

- Powers the S32K144EVB evaluation board from a USB. By default, the USB power is enabled by J107 jumper (2-3 closed).
- Connect the USB cable to a PC and connect micro USB connector of the USB cable to micro-B port J7 on the S32K144EVB.
- Allow the PC to automatically configure the USB drivers if needed.
- When EVB is powered from USB, LEDs D2 and D3 should light green.
- The EVB board is preloaded with a software toggling the RGB LED colours periodically between RED-GREEN-BLUE.

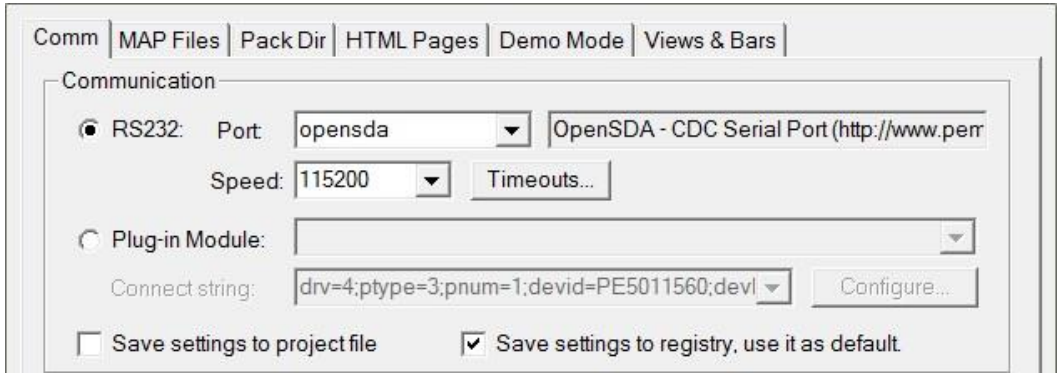




Setup serial connection in the FreeMASTER tool

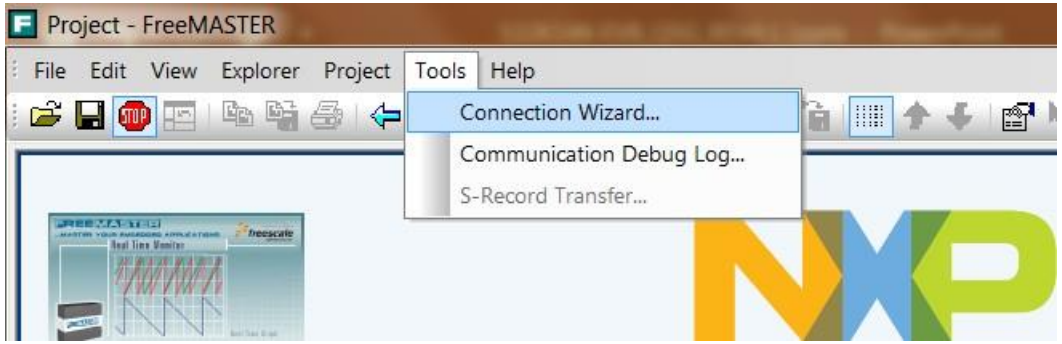
Setup communication port to „opensda“ and speed to 115200 b/s:

- Setup communication manualy: „Project > Options > Comm“



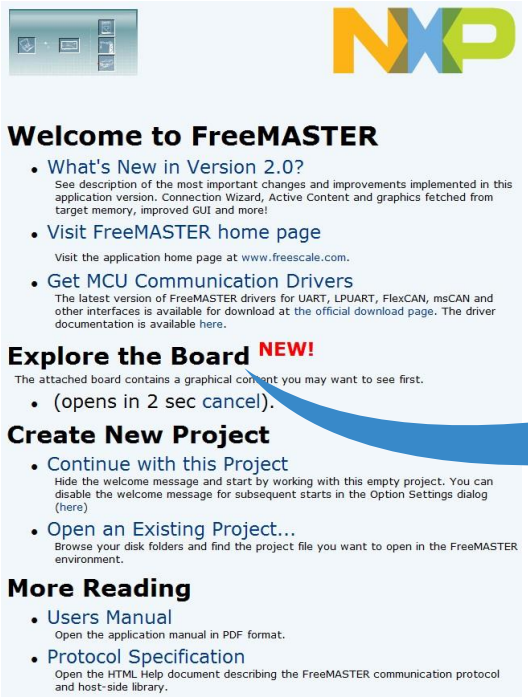
OR

- Setup communication automatically: „Tools > Connection Wizard“

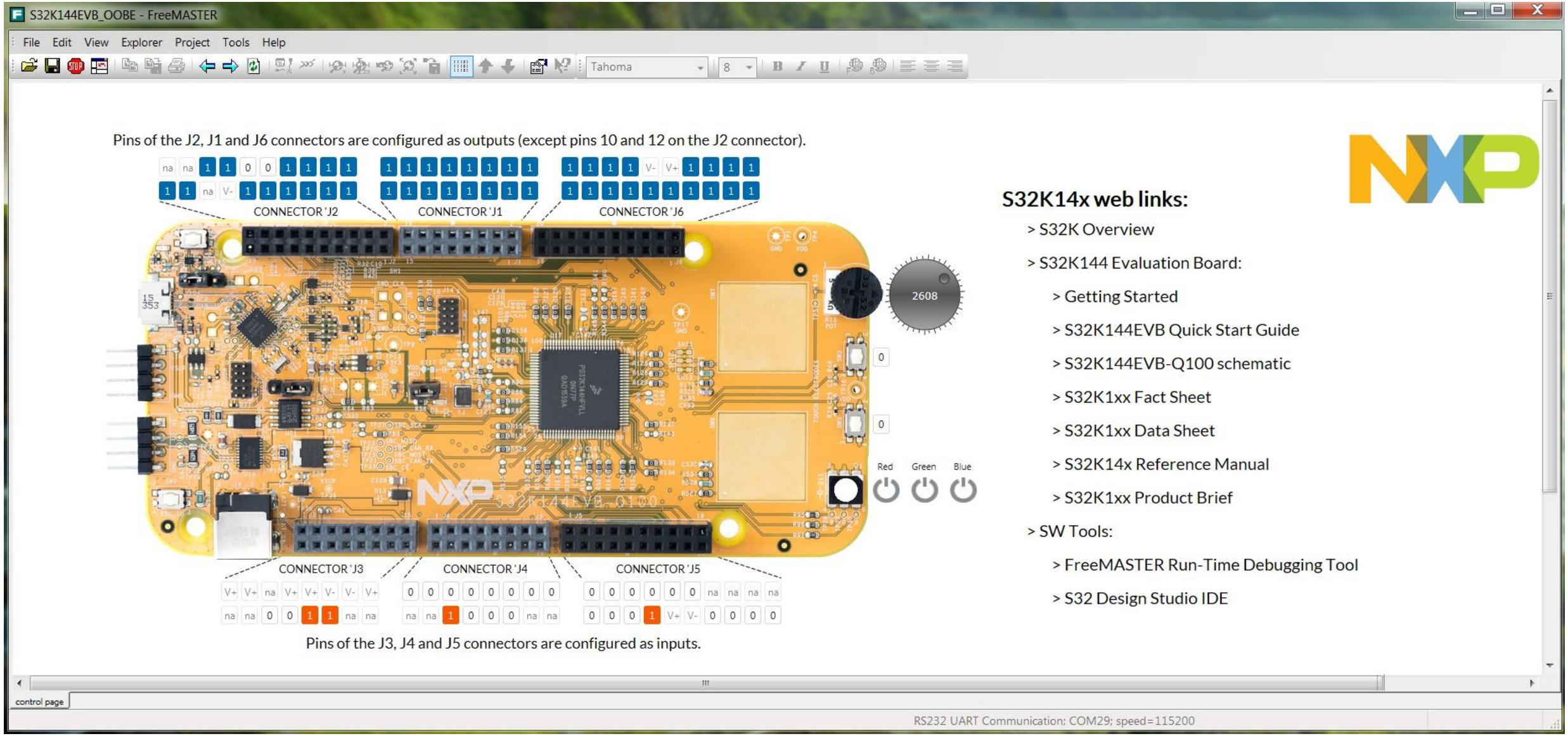


The FreeMASTER JumpStart project will be automaticaly downloaded from www.nxp.com

Once the FreeMASTER application detects the web address stored as an TSA active content in the flash memory of the S32K144 MCU, the download of the FreeMASTER project from www.nxp.com will be initiated.

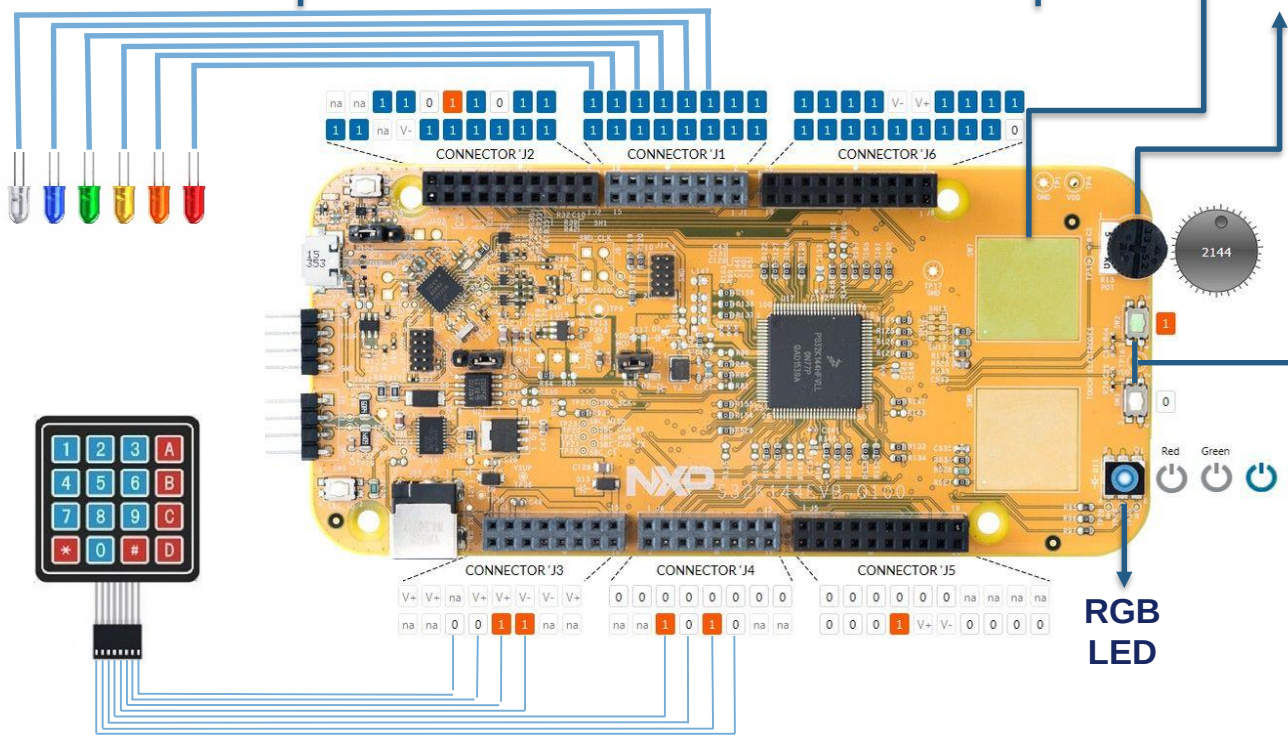


The FreeMASTER JumpStart project is loaded



The FreeMASTER JumpStart project description

Pins of the J2, J1 and J6 connectors are configured as outputs. By single click on each pin you can change their logical level to log0 or log1. User can connect e.g. LED diodes to these ouput pins.



Touch Sense Electrodes

Potentiometer

RGB LED

Mechanical Buttons

S32K14x Web Links:

- >> S32K Overview
- >> S32K144 Evaluation Board:
  - > Getting Started
  - > S32K144EVB Quick Start Guide
  - > S32K144EVB-Q100 Schematic
  - > S32K1xx Fact Sheet
  - > S32K1xx Data Sheet
  - > S32K14x Reference Manual
  - > S32K1xx Product Brief
- >> SW Tools:
  - > FreeMASTER Run-Time Debugging Tool
  - > S32 Design Studio IDE
- >> S32K144 EVB JumpStart Sources:
  - > S32K144 EVB JumpStart PC Host Project
  - > S32K144 EVB JumpStart Firmware

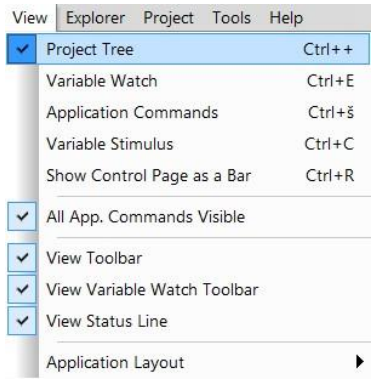
Links to S32K14x docs:

- **Fact Sheet**
- **Data Sheet**
- **Reference Manual**
- **Product Brief**
- **S32K144EVB schematic**
- **S32K144EVB Quick Start Guide**
- **Tools:**
  - **FreeMASTER**
  - **S32 Design Studio IDE**
- **S32K144EVB OOBE source files**

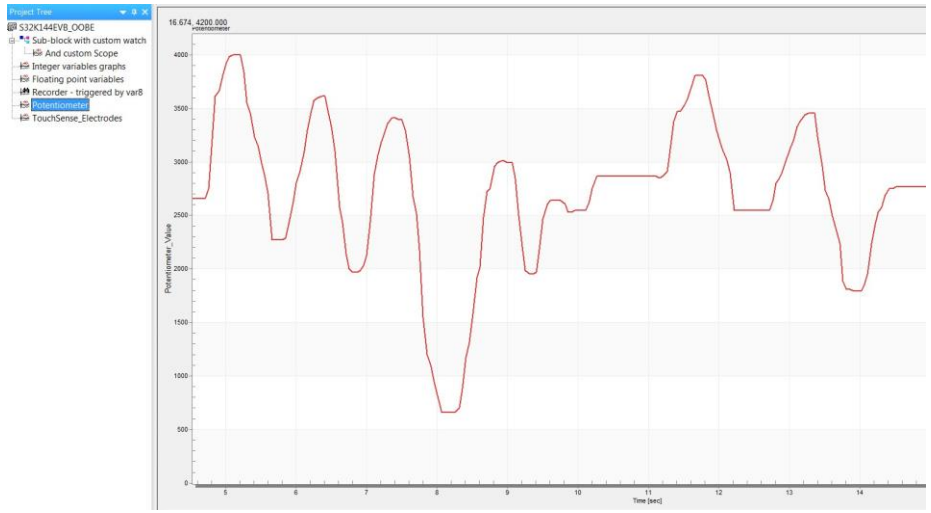
Pins of the J3, J4 and J5 connectors are configured as inputs. Logical level (log0/log1) is visualised for all connector pins. User can connect e.g. push-button keyboard to these input pins.

The FreeMASTER JumpStart oscilloscope feature examples

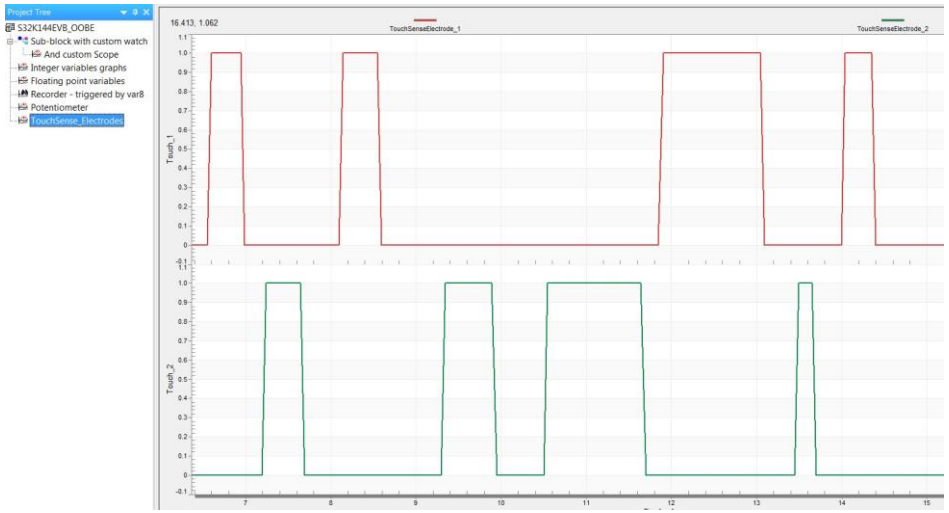
- Display main project panel „Project > View > Project Tree“.



- Display real-time oscilloscope graph examples such as „Potentiometer“ or „Touch Sense Electrodes“.



Analog values from potentiometer.



Responses from touch sense electrodes.

INTRODUCTION TO OPENSDA

Introduction to OpenSDA: 1 of 2

OpenSDA is an open-standard serial and debug adapter. It bridges serial and debug communications between a USB host and an embedded target processor. OpenSDA software includes a flash-resident USB mass-storage device (MSD) bootloader and a collection of OpenSDA Applications. S32K144 EVB comes with the MSD Flash Programmer OpenSDA Application preinstalled. Follow these instructions to run the OpenSDA Bootloader and update or change the installed OpenSDA Application.

Enter OpenSDA Bootloader Mode

1. Unplug the USB cable if attached
2. Set J104 on position 1-2.
3. Press and hold the Reset button (SW5)
4. Plug in a USB cable (not included) between a USB host and the OpenSDA USB connector (labeled “SDA”)
5. Release the Reset button

A removable drive should now be visible in the host file system with a volume label of BOOTLOADER. You are now in OpenSDA Bootloader mode.

IMPORTANT NOTE: Follow the “Load an OpenSDA Application” instructions to update the MSD Flash Programmer on your S32K144 EVB to the latest version.

Load an OpenSDA Application

1. While in OpenSDA Bootloader mode, double-click ***SDA_INFO.HTML*** in the ***BOOTLOADER*** drive. A web browser will open the OpenSDA homepage containing the name and version of the installed Application. This information can also be read as text directly from ***SDA_INFO.HTML***
2. Locate the ***OpenSDA Applications***
3. Copy & paste or drag & drop the MSD Flash Programmer Application *to the* ***BOOTLOADER*** *drive*
4. Unplug the USB cable and plug it in again. The new OpenSDA Application should now be running and a ***S32K144 EVB*** drive should be visible in the host file system

You are now running the latest version of the MSD Flash Programmer. Use this same procedure to load other OpenSDA Applications.

Introduction to OpenSDA: 2 of 2

The MSD Flash Programmer is a composite USB application that provides a virtual serial port and an easy and convenient way to program applications into the KEA MCU. It emulates a FAT16 file system, appearing as a removable drive in the host file system with a volume label of EVB-S32K144. Raw binary and Motorola S-record files that are copied to the drive are programmed directly into the flash of the KEA and executed automatically. The virtual serial port enumerates as a standard serial port device that can be opened with standard serial terminal applications.

Using the MSD Flash Programmer

1. Locate the .srec file of your project , file is under the Debug folder of the S32DS project.
2. Copy & paste or drag & drop one of the .srec files to the EVB-S32K144 drive

The new application should now be running on the S32K144 EVB. Starting with v1.03 of the MSD Flash Programmer, you can program repeatedly without the need to unplug and reattach the USB cable before reprogramming.

Drag one of the .srec code for the S32K144 the S32K144 EVB board over USB to reprogram the preloaded code example to another example.

NOTE: Flash programming with the MSD Flash Programmer is currently only supported on Windows operating systems. However, the virtual serial port has been successfully tested on Windows, Linux and Mac operating systems.

Using the Virtual Serial Port

1. Determine the symbolic name assigned to the EVB-S32K144 virtual serial port. In Windows open Device Manager and look for the COM port named “PEMicro/Freescale – CDC Serial Port”.
2. Open the serial terminal emulation program of your choice. Examples for Windows include Tera Term, PuTTY, and HyperTerminal
3. Press and release the Reset button (SW0) at anytime to restart the example application. Resetting the embedded application will not affect the connection of the virtual serial port to the terminal program.
4. It is possible to debug and communicate with the serial port at the same time, no need to stop the debug.

NOTE: Refer to the OpenSDA User’s Guide for a description of a known Windows issue when disconnecting a virtual serial port while the COM port is in use.

INSTALLING S32DS

Download S32DS

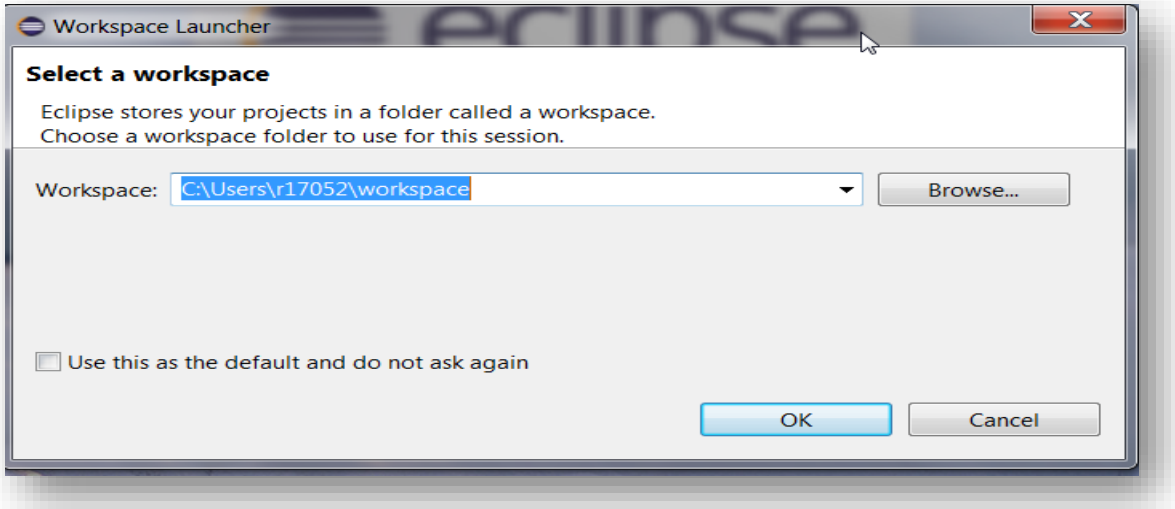
Download S32DS from:

http://www.nxp.com/S32DS

CREATE A NEW PROJECT IN S32 DESIGN STUDIO

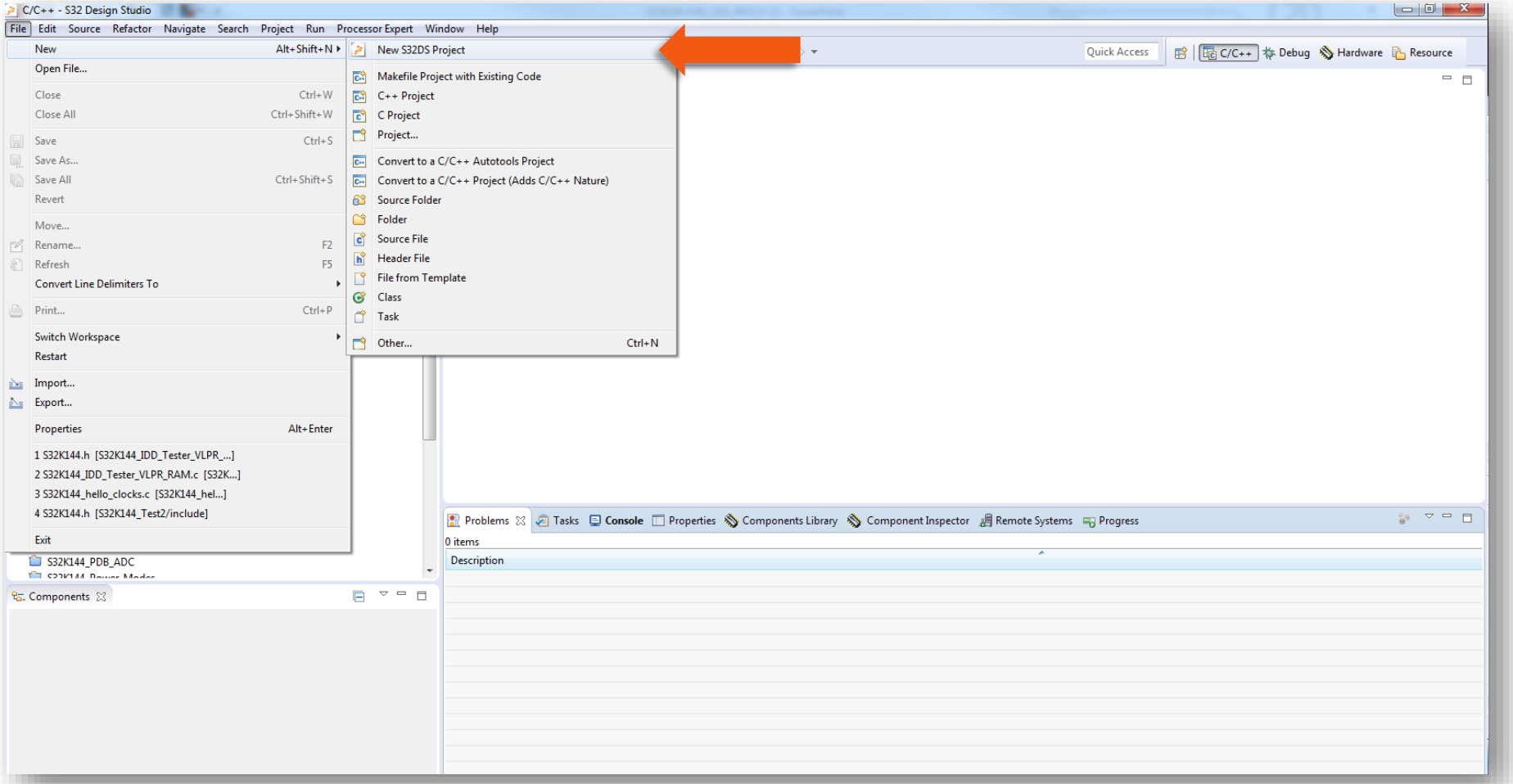
Create New Project: First Time – Select a Workspace

- Start program: Click on “S32 Design Studio for ARM v1.3” icon
- Select workspace:
  - Choose default (see below example) or specify new one
  - Suggestion: Uncheck the box “Use this as the default and do not ask again”
  - Click OK



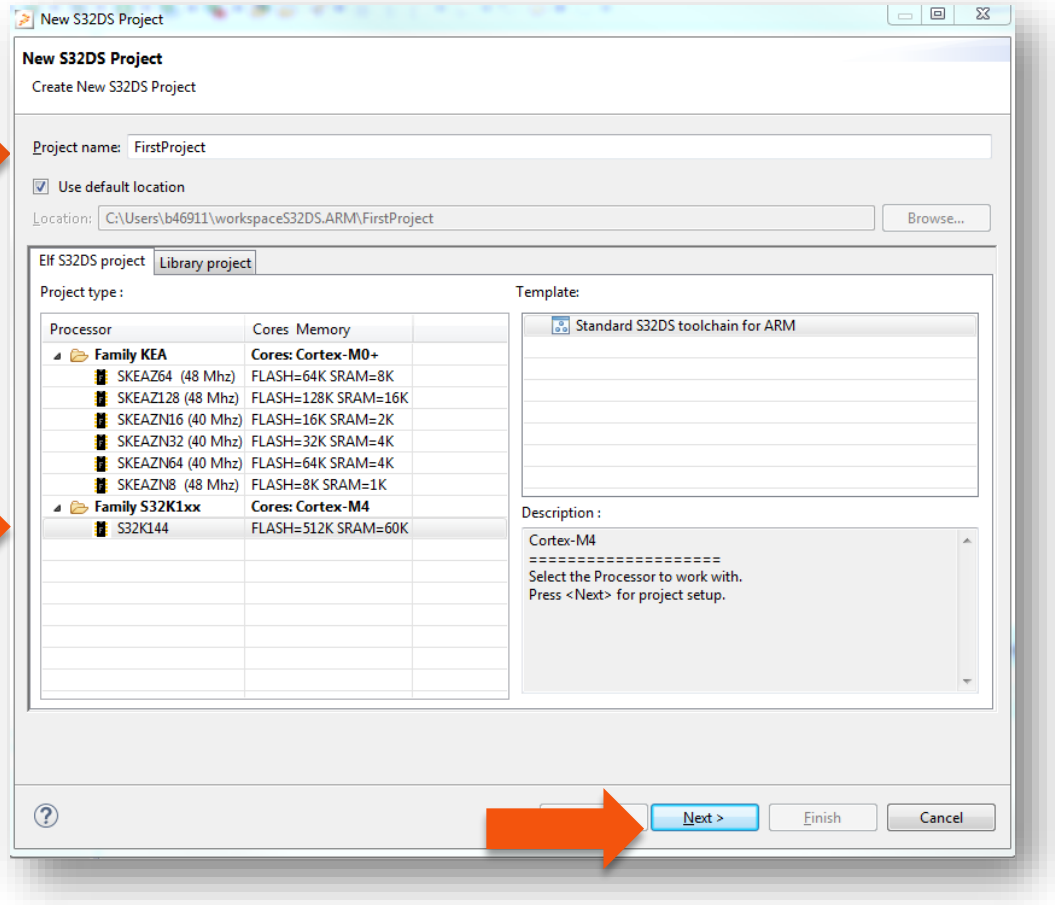
Create New Project: Top Menu Selection

- File – New –Project



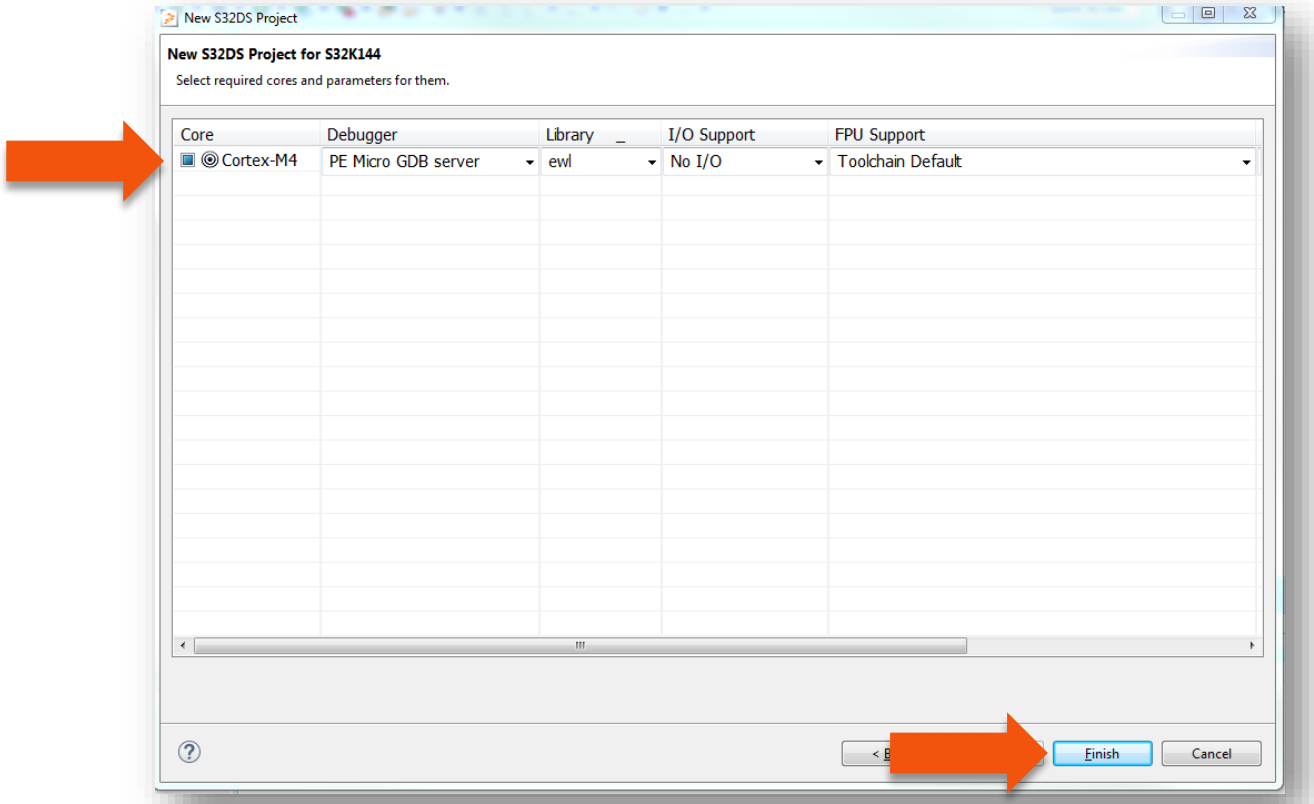
Create New Project: S32DS Project

- Project Name:
  - Example: FirstProject
- Project Type:
  - Select from inside executable or library folder
- Next



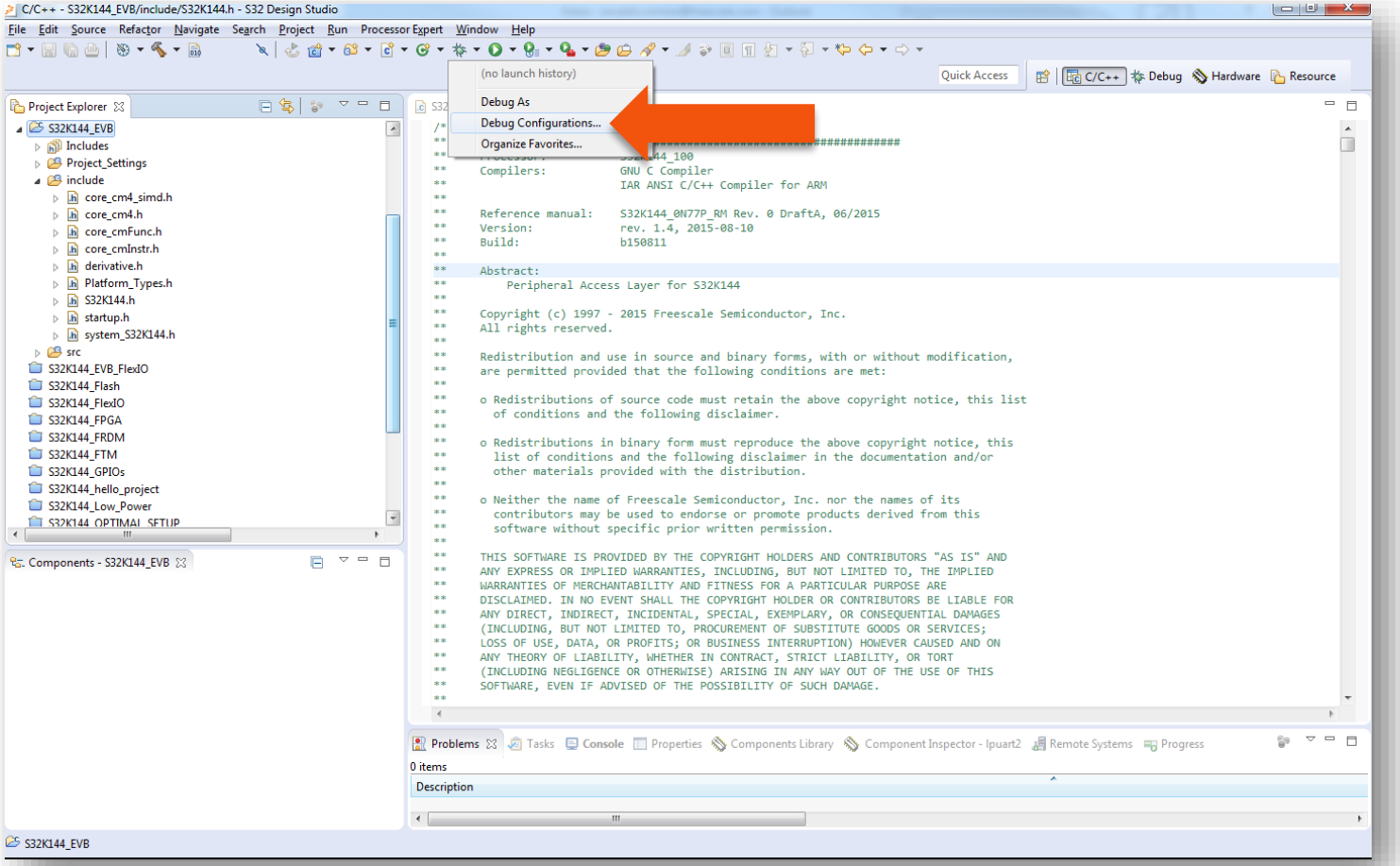
Create New Project: S32DS Project

- Select Debugger Support and Library Support
- Click Finish



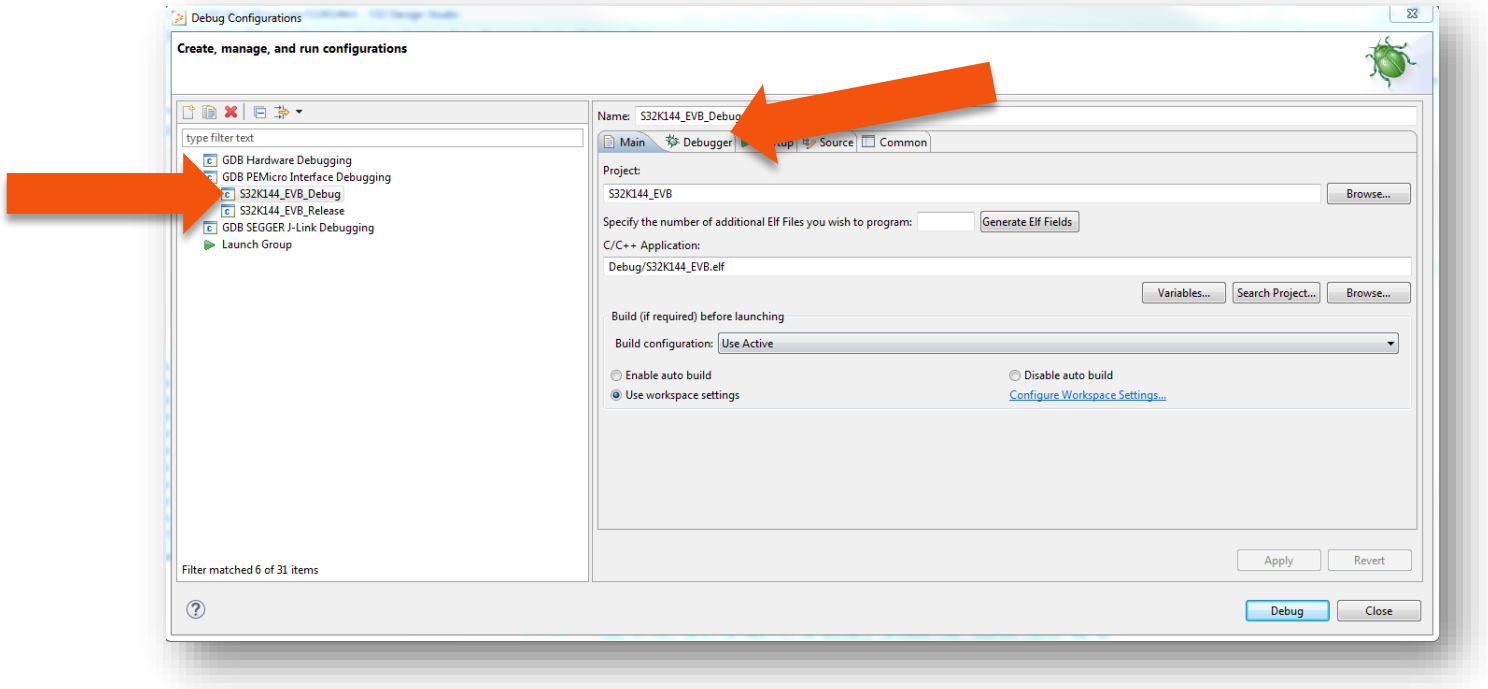
OpenSDA Configuration

- To Debug your project with OpenSDA, it is necessary to select the OpenSDA in the Debug Configuration.
- Select your project, and click on debug configuration



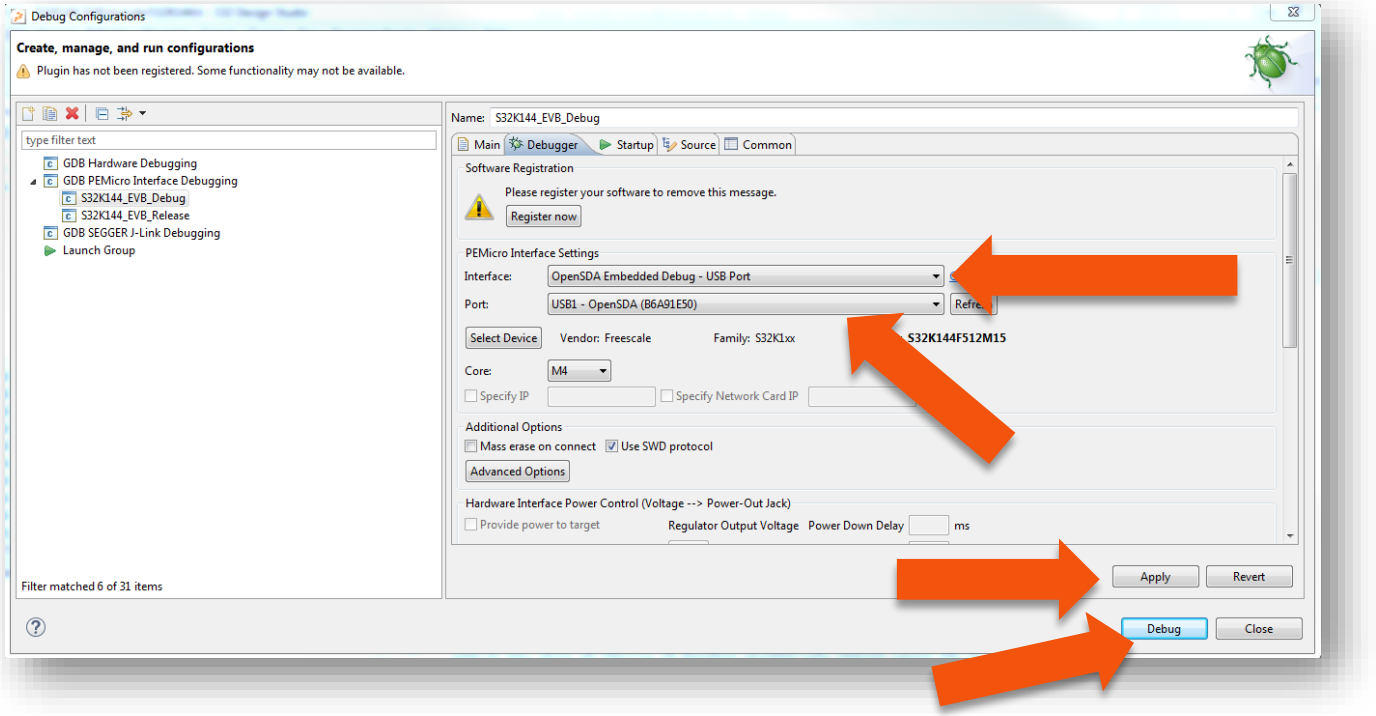
OpenSDA Configuration

- Select the Debug configuration under GDB PEMicro Interface Debugging
- Click on Debugger tab



OpenSDA Configuration

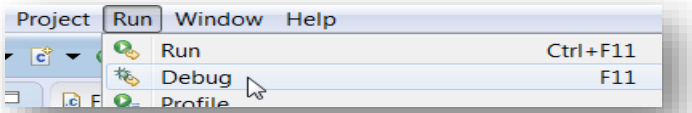
- Select OpenSDA as the interface, if your board is plugged should appear in the Port field.
- Click Apply and debug to finish.



DEBUG BASICS

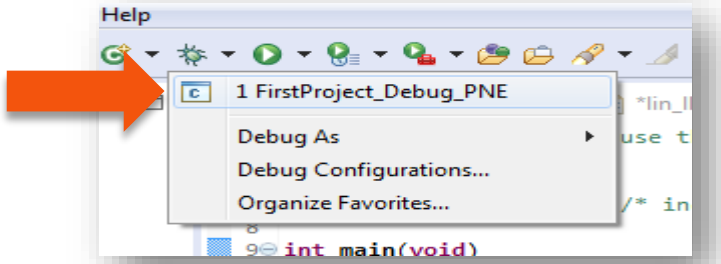
Debug Basics: Starting the Debugger

- Debug configuration is only required once. Subsequent starting of debugger does not require those steps.
- Three options to start debugger:
  - If the “Debug Configuration” has not been closed, click on “Debug” button on bottom right
  - Select Run – Debug (or hit F11)



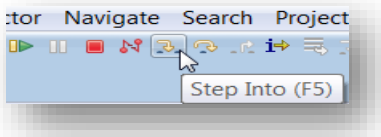
Note: This method currently selects the desktop target (*project.*elf) and gives an error. Do not use until this is changed.

  - *<u>Recommended Method</u>*: Click on pull down arrow for bug icon and select …_debug.elf target

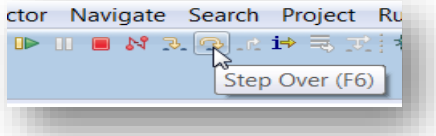


Debug Basics: Step, Run, Suspend, Resume

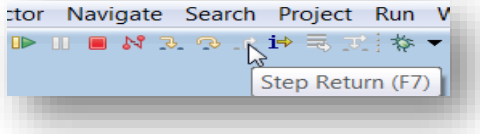
- Step Into (F5)



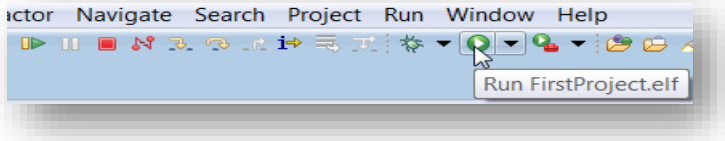
- Step Over (F6)



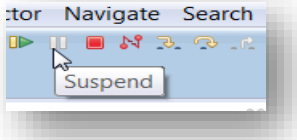
- Step Return (F7)



- Run



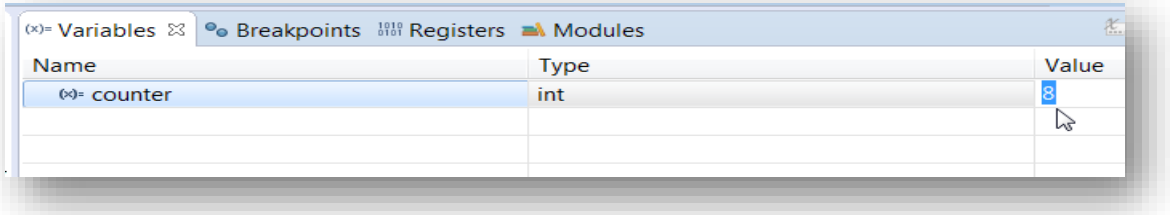
- Suspend



- Resume (F8)

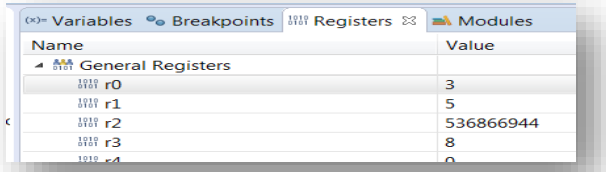
Debug Basics: View & Alter Variables

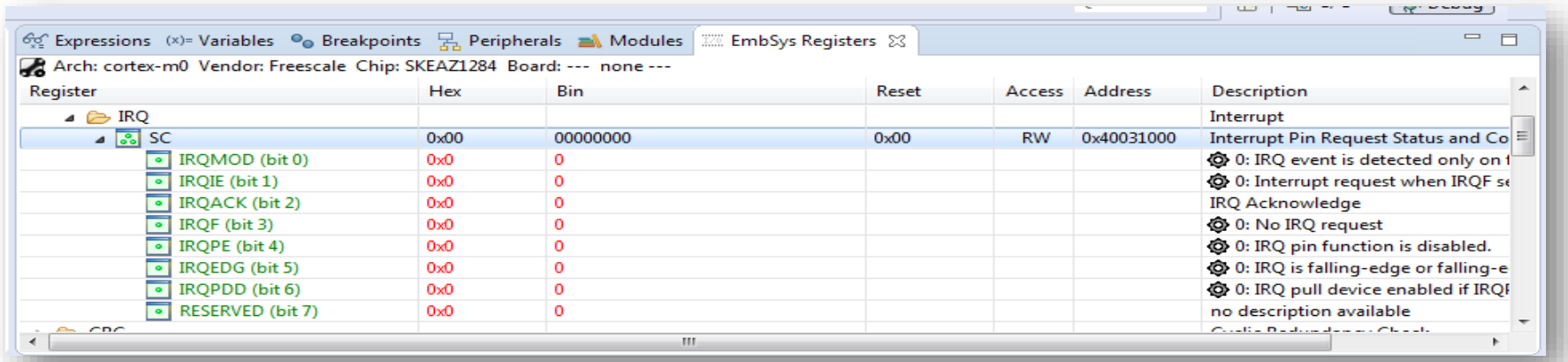
- View variables in “Variables” tab.
- Click on a value to allow typing in a different value.



Debug Basics: View & Alter Registers

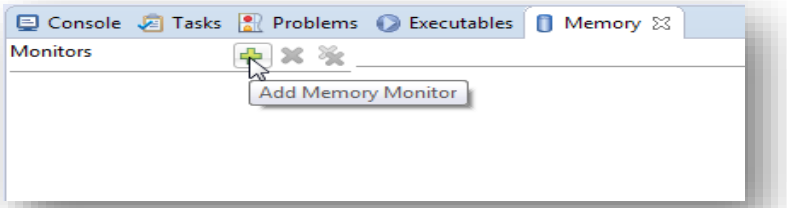
- View CPU registers in the “Registers” tab
- Click on a value to allow typing in a different value
- View peripheral registers in the EmbSys Registers tab



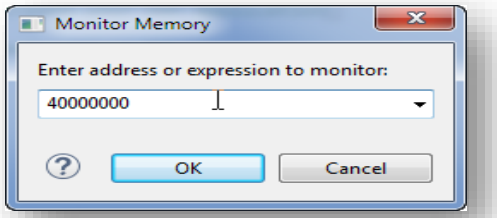


Debug Basics: View & Alter Memory

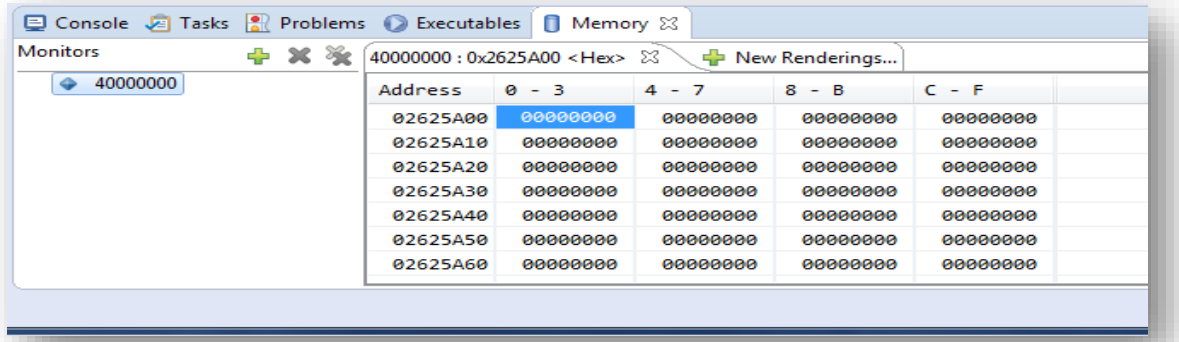
- Add Memory Monitor



- Select Base Address to Start at : 40000000



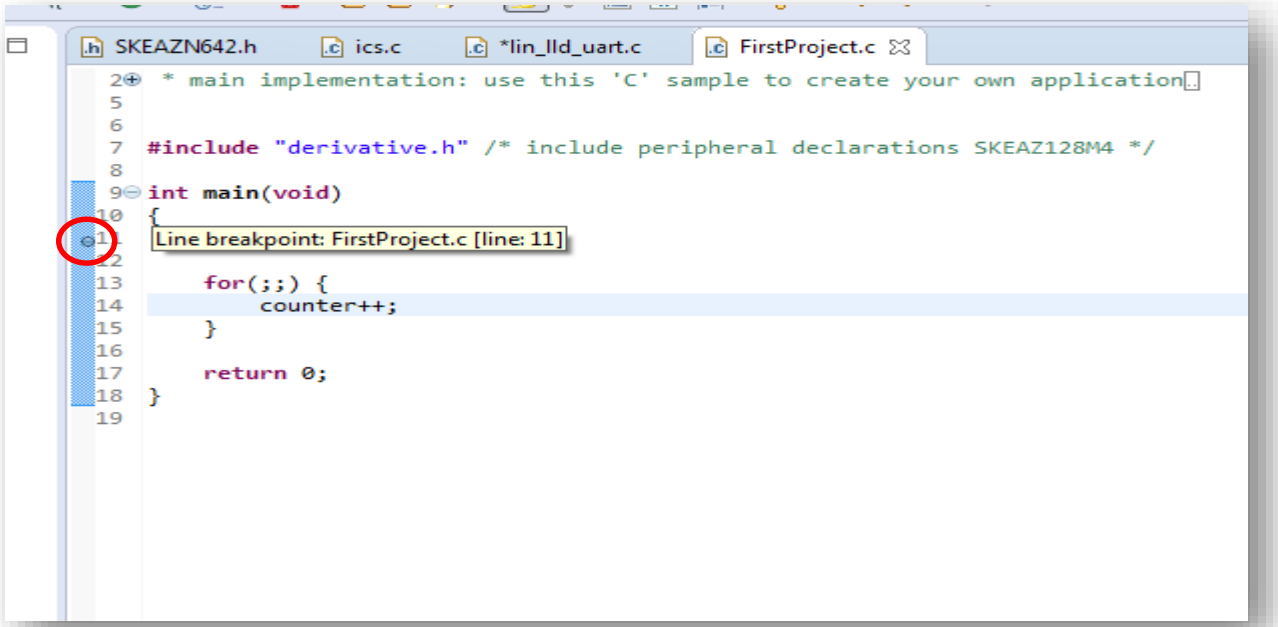
- View Memory



Debug Basics: Breakpoints

Add Breakpoint: Point and Click

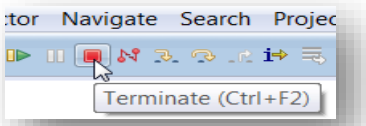
- light blue dot represents debugger breakpoint



Debug Basics: Reset & Terminate Debug Session

- Reset program counter

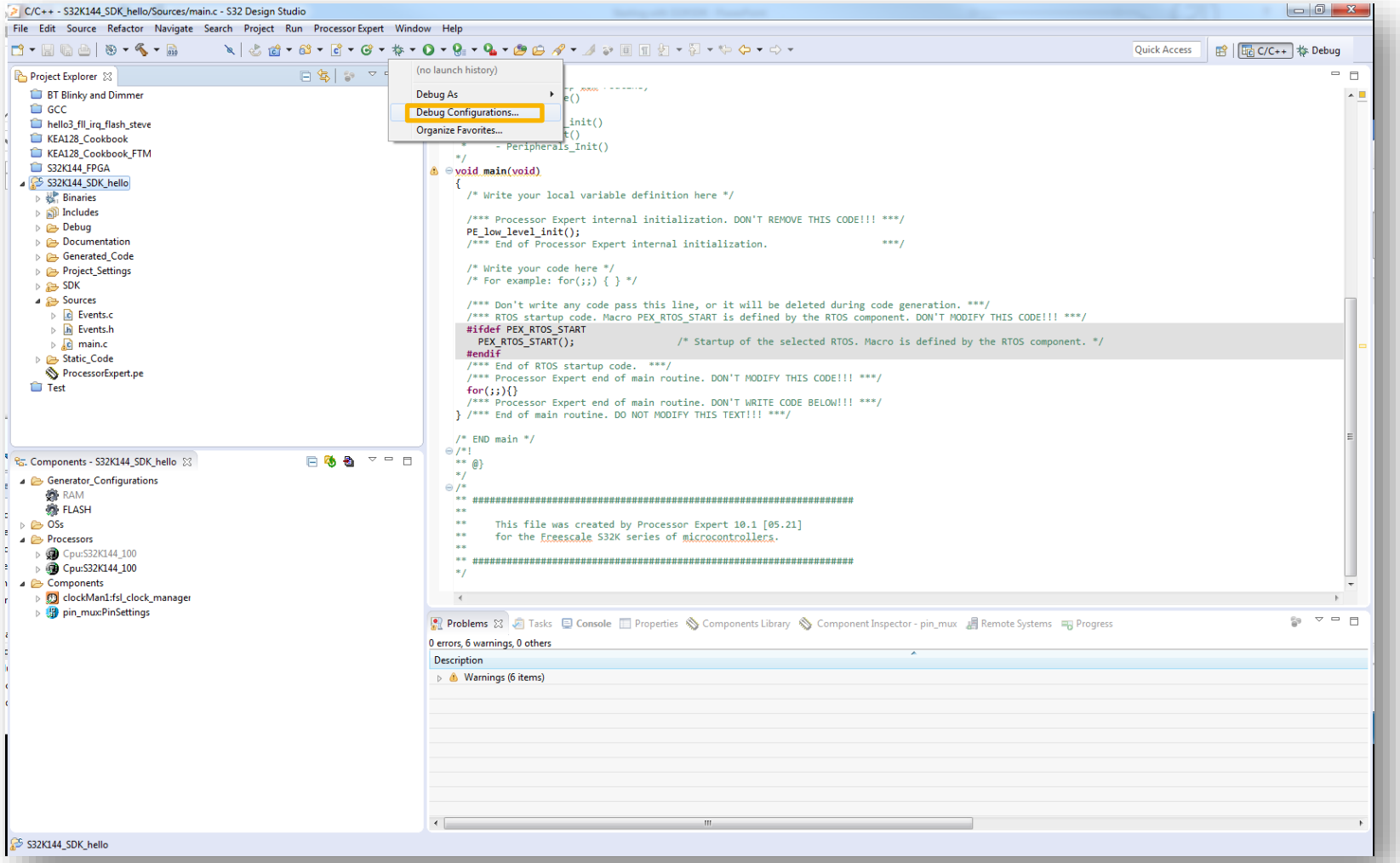
- Terminate Ctl+F2()



CREATE A P&E DEBUG CONFIGURATION (OPTIONAL)

New P&E debug configuration

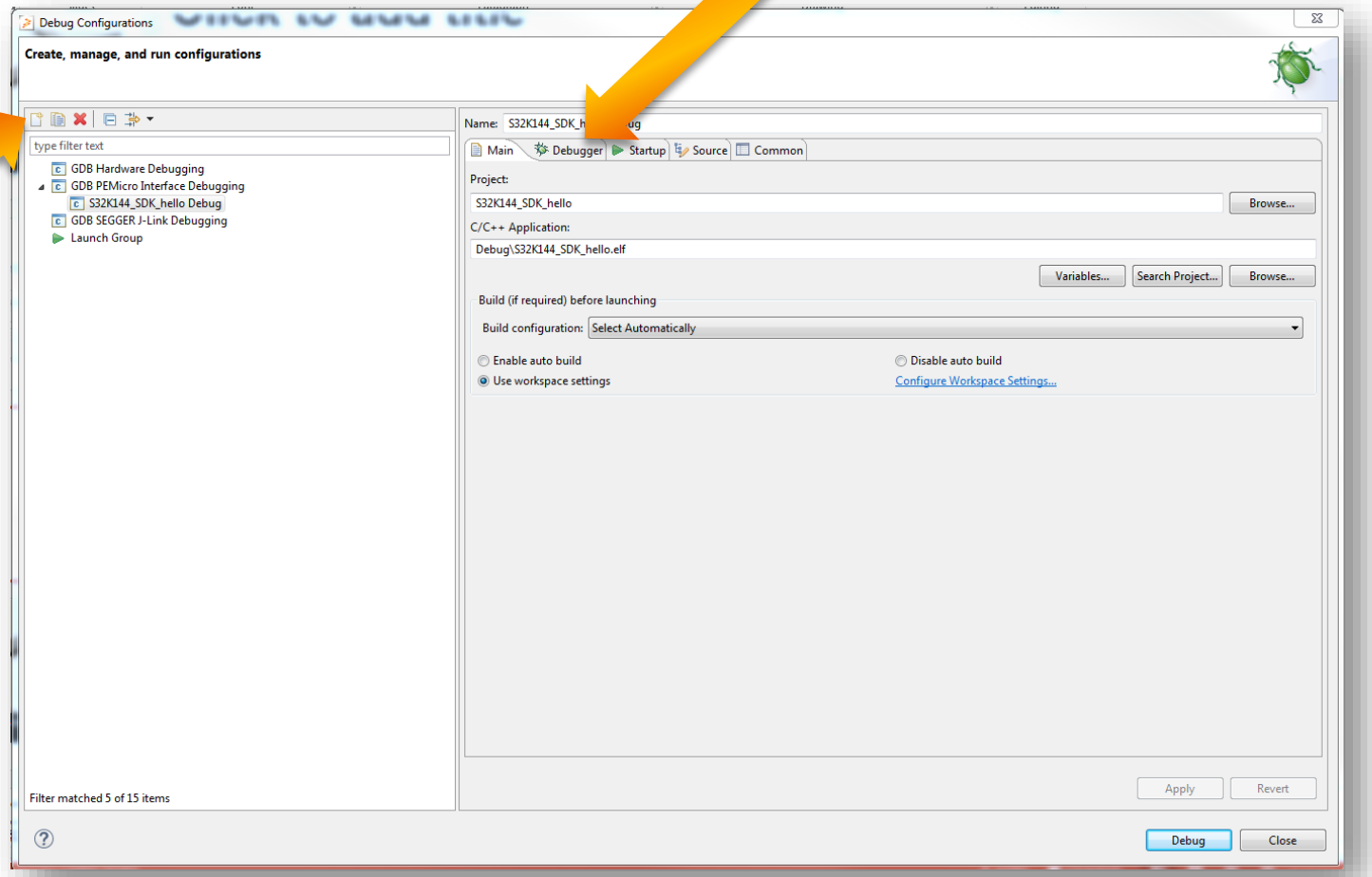
- Click in debug configurations



New P&E debug configuration

- Create a new P&E launch configuration

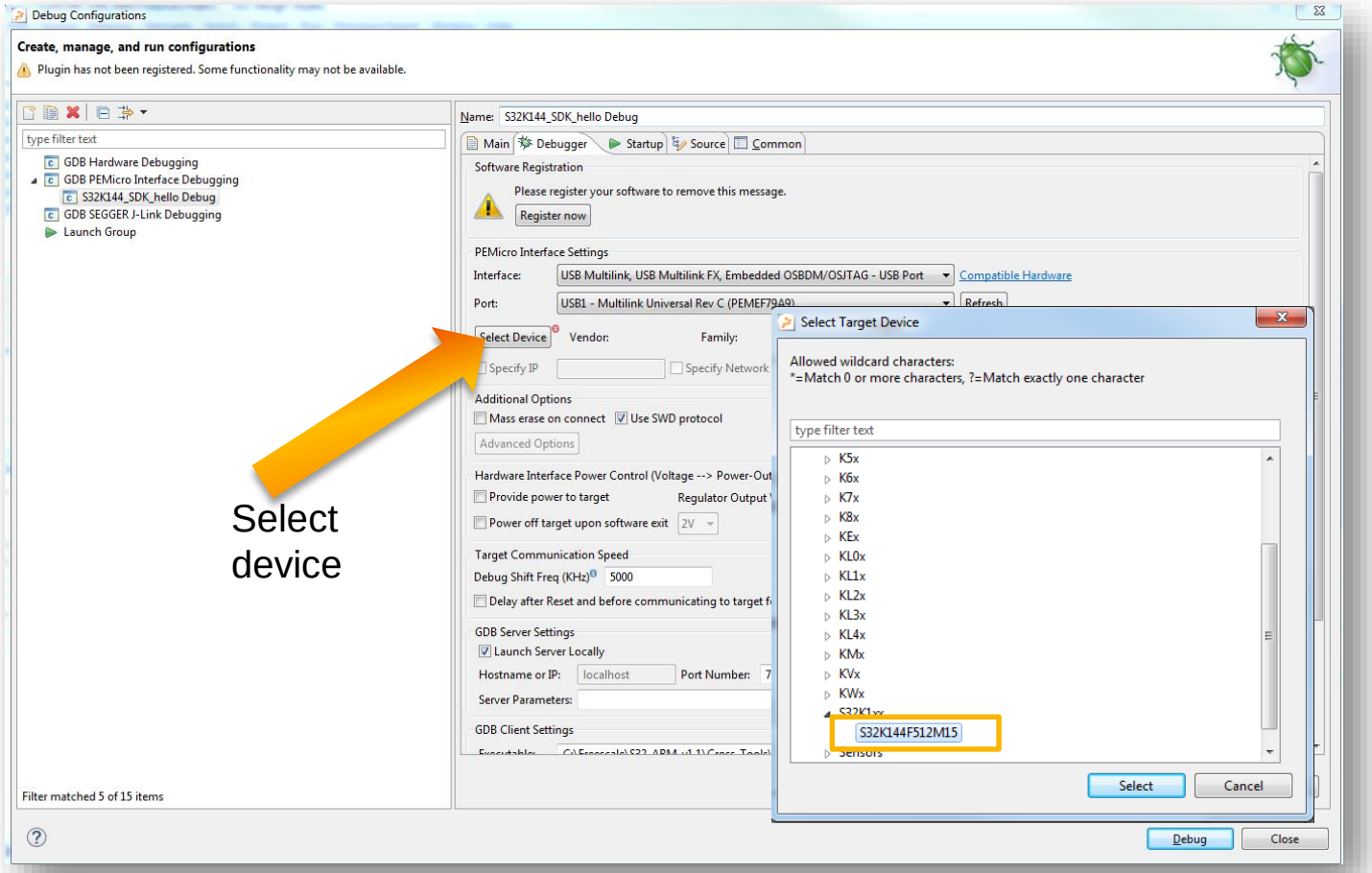
Click on the debugger tab.



Click to create a new P&E launch

New P&E debug configuration

- Select S32K144 device



Select device

- Click Apply and debug your application

SECURE CONNECTIONS
FOR A SMARTER WORLD